

MODULES

Module Documentation

This document provides detailed documentation for each module in the Oro Ticket App, including business logic explanations and implementation details.

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Authentication Module

Overview

The authentication module handles user login, session management, and secure token storage.

Structure

Business Logic

Login Flow

- 1. \$1
 - Email format validation
 - Password presence check
 - Form state management
 - 2. \$1
 - 3. \$1
 - JWT token storage in secure storage
 - Automatic token refresh
 - Session timeout handling
- Offline Authentication
- \$1: Cached credentials for offline login
 - \$1: Local token expiry checks
 - \$1: Background sync after reconnection

Key Components

AuthService

\$1
API communication for authentication
Token management
Profile data synchronization
Logout handling with data validation

\$1

SignInController

\$1

Password visibility toggle
Form validation
Error handling and user feedback
Navigation after successful login

Home/Dashboard Module

Overview

The home module serves as the main dashboard displaying key metrics and providing navigation to all features.

Structure

Business Logic

Dashboard Metrics

- 1. \$1
- 4. \$1
Unique ticket numbering
Ethiopian date/time formatting
QR code generation for verification
Company branding inclusion
- Printing Integration
\$1: Automatic printer detection
\$1: Thermal printer-specific formatting
\$1: Print failure recovery
\$1: Digital ticket storage

Key Components

Ticket Data Structure

Business Rules
\$1: Cannot exceed vehicle seat capacity
\$1: Only predefined routes allowed
\$1: Tickets valid for scheduled times
\$1: Unique ticket identification

Trip Management Module

Overview

The trip module manages trip recording, synchronization, and historical data.

Structure

Business Logic

Trip Recording

1. \$1

Vehicle and route information

Passenger count and revenue

Actual distance traveled

Timestamp recording

2. \$1

3. \$1

Data Validation

\$1: All required fields present

\$1: Distance and time reasonable

\$1: Prevent duplicate trip entries

\$1: Revenue matches expected calculations

Key Features

Filtering and Search

\$1

\$1

\$1

\$1

Bulk Operations

\$1

\$1

\$1

Fleet Management Module

Overview

The fleet module manages vehicle information, assignments, and maintenance tracking.

Structure

Business Logic

Vehicle Data Management

1. \$1

2. \$1

Departure terminal association

Arrival terminal configuration

Tariff rate assignment

3. \$1

- Active/Inactive status
- Maintenance scheduling
- Assignment tracking

Data Synchronization

- \$1: Only changed data synced
- \$1: Server data takes precedence
- \$1: Full vehicle data available offline

Key Features

Vehicle Search and Filter

- \$1
- \$1
- \$1
- \$1

Assignment Management

- \$1
- \$1
- \$1

Reporting Module

Overview

The reporting module provides comprehensive reporting capabilities with PDF generation.

Structure

Business Logic

Report Generation

- 1. \$1
- 2. \$1

Report Types

- 1. \$1
 - Daily trip summaries
 - Revenue analysis
 - Route performance
- 2. \$1
 - Commission tracking
 - Employee performance
 - Financial summaries
- 3. \$1
 - Vehicle utilization
 - Maintenance schedules
 - Fuel efficiency

Key Features

Filtering Capabilities

- \$1: Custom date selection
- \$1: Specific vehicle reports
- \$1: Terminal-based filtering
- \$1: Revenue-based filtering

Export Options

- \$1: Formatted reports
- \$1: Raw data export
- \$1: Direct report distribution

Synchronization Module

Overview

The synchronization module manages data synchronization between local storage and remote servers.

Structure

Business Logic

Synchronization Strategy

1. \$1

2. \$1

Only sync changed data

Version-based conflict resolution

Efficient bandwidth usage

3. \$1

Data Types Synchronized

- \$1: Vehicles, terminals, commission rules
- \$1: Trips, service charges, tickets
- \$1: User profiles, company settings

Key Features

Progress Tracking

- \$1: Synchronization progress indicators
- \$1: Detailed error messages
- \$1: Automatic retry for failed operations

Conflict Resolution

- \$1: Server data takes precedence
- \$1: User intervention for conflicts
- \$1: Synchronization history

Settings Module

Overview

The settings module manages application configuration and user preferences.

Business Logic

Configuration Management

1. \$1
API endpoint configuration
Debug mode settings
Feature flags

2. \$1
Language selection
Theme preferences
Notification settings

3. \$1
Printer configuration
Storage management
Permission management

Key Features

App Configuration
\$1: App version and build details
\$1: Cache clearing and data management
\$1: Device permission management

User Management
\$1: User information updates
\$1: Password change functionality
\$1: Logout and session control

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\$1: November 2024
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